TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Solve: $x = y + 2$ and $x + y = 12$.

Tip: substitute $y + 2$ for x

A. (5, 7)

B. (7, 5)

C. (6, 6)

2. What is the additive inverse of $5y$?

Tip: opposite sign, same term

F. $5y$

G. $\frac{1}{5}y$

H. $-5y$

3. Add the equations $3x + 2y = 7$ and $x - 2y = 5$. What do you get?

> **Tip:** add like terms down the columns

A. $4x - 4y = 12$

B. $2x = 2$

C. $4x = 12$

4. Multiply every term of $2x - y = 3$ by 3.

> **Tip:** multiply EVERY term

F. $6x - 3y = 9$

G. $6x - y = 3$

H. $6x - 3y = 3$